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# Lab Final Project Report: Temperature Sensor

# Introduction

For EE330, the last task of the students was to pick a lab project from a select group and design a schematic and Layout of the electronic device. Once the students complete this task, they are to demo their work and present their project to a TA and finally write a report of the overall process of the assigned task. This is a group effort assignment between two students. This report was written in collaboration between Eric Williams and Connor Crimmins on a joint effort to build the project they were assigned to. The project we were assigned to do is the Temperature Sensor Project. This will be shown in more detail in our Background. The following document will show the reader how we were able to create the Temperature Sensor for both schematic and layout design as well as the struggles they faced during the project. In conclusion, the reader will be able to note an overall reflection of the processes taken to solve the task as well as a conclusion to EE330: Integrated Electronics as a result.

# Background

## Temperature Sensor

The assigned task of this project was for us to successfully create a temperature sensor device using the IC design technical skills that were gained from the knowledge introduced in this course. In the figure below we can see an overview of what was asked of the students who were assigned to the Temperature Sensor Project:

### Temperature Sensor

Temperature sensors have a wide range of applications. This project will focus on the design of an on-chip temperature sensor that provides an analog output that varies linearly with absolute temperature (PTAT). It should operate over the range of 0C to 50C with an output of 0V at 0C and an output of 1V at 50oC. A pair of diodes will be used as a temperature sensor. The project will also likely be easier to complete if an operational amplifier is used. Since the topic of designing operational amplifiers is beyond the scope of this course, a method will be provided for designing an externally compensated operational amplifier for this project. Though we have not discussed temperature sensors in class, a 1-hour discussion with the course instructor or the TA should be sufficient to obtain the understanding needed to build this temperature sensor

As the reader can see, the goal was to make a PTAT Temperature Sensor. A PTAT Temperature Sensor stands for Proportional To Absolute Temperature. All this means that the temperature sensor should be able to have a linear relationship between voltage and temperature. As temperature increases, this should linearly be proportional to voltage levels that can be an analog input to the sensing device. This will in return create a system

that will read the temperature of a peripheral device and depend on the application will be able to communicate with other peripheral devices to help solve the temperature issues. With this project, the students were tasked with solving a sensor that would read about a range of temperatures from 0C to 50C. This will be linearly proportional to an analog voltage of 0V to 1V. The lab document encourages student participants to design an operational amplifier to help with the processing of this signal. This topic is beyond the scope of this course, so conversations with TAs as well as personal research on the matter were needed to complete the project.

For this project to work, our TA (Sam Burns) informed us to start by doing our own research on Diodes and how they are temperature dependent. To do this properly, Connor investigated Science Direct to find research papers regarding temperature sensors. Eric was advised by Sam to look into it by using ChatGPT because it is faster for us to understand the information and because the TA (Sam) said that Dr. Geiger approved the use of AI. In our findings, we found that through the Diode equation, although this is a non-linear equation, the diode equation is temperature dependent. The concept with the help of AI was found was that the saturation current is also temperature dependent, which would help the students understand why this circuit is allowed to work. After we looked into this, we decided to investigate the Operational Amplifier.

## Operational Amplifier Findings

For our research on op amps, we looked back at previous year's documents for when they were in EE230: Electronic Circuits and Systems. Our reasoning at the time was strict because they had built a Temperature Sensor before in that course. They decided the best course of action was to use a differential op amp. The reasoning is that we thought it would be a good idea since the differential op amp takes the difference between the two signals at the Voltage inputs of the amplifier. When doing so, the user of the op amp can directly take that difference and use it to determine whether or not that the difference is high or low and take that information and convert it to temperature readings which would in return create a sufficient Temperature Sensor.

When looking through documents, we found that there were two op amps steadily used for Temperature Sensor devices; Comparators and Differential op amps. A Differential op amp differs from a Comparator because rather than comparing for which is higher in value, a Differential op amp keeps the system in the linear region and takes the difference between the two input signals. This is something that is good for the circuit because it informs the user that we will stay in linear mode, which is something that we did not

remember at first.  To test this op amp we had to do a couple of tests to confirm it works. Firstly, we had to create a test bench for the op amp and confirm that it works. We did this by using the equations and theories about transistor design that they learned in class to derive parameters for the transistors being used in the design. For this project, it was very difficult to figure out an efficient way to do this. Dr. Geiger gives an Excel sheet that you can use to extract your parameters for the op amp. However, what we found out a little late for the project was that we didn't need to change much about the excel document. All we really had to do was use the excel sheet we made to extract the parameters in Lab 7 and then use them to help solve the excel documents. The reader can see the excel document in more detail in the figure below. The op amp circuit was created by looking over Dr. Geiger's Home Page which shows a document about Differential op amps that gives a 7T design.  Once you create the circuit, you can create a symbol for it for the next step. Next, a testbench for its circuit that is comprised of resistors that were used to create a differential op amp as well as a capacitor for stability of the phase margin. Once the testbench is successfully created, we can do DC Analysis on the testbench to see a linear graph that represents the voltage gain of the op amp.  Finally, before being able to use the op amp in schematic and layout, we need to confirm the phase margin. This can be done by using I-PROBE on the feedback line to test for phase. The output of this shows that the op amp is stable, but it may have some errors with it due to limitations we did not know were present at the time. When building this circuit, we did not take this into account, however we still ended up with a correct output on our schematics and used this op amp for the duration of the project.

# Operational Amplifier

## Schematic

To create the op amp, we were given a document and spreadsheet from the 330 websites to assist us since the gpdk045 process did not have any nonideal amplifiers to use. The document gave us the design for a 7T amplifier and some information on how to use the datasheet. For this setup, we decided to use NMOS2V and PMOS2V transistors since those would give us enough room to get to an output of 1V without clipping. The 2V means that these transistors can't go above 2V of input magnitude, which means that our VDD and VSS had to be below this mark. We set VDD to be 1.8V and VSS to be -1.8V to adhere to this rule.

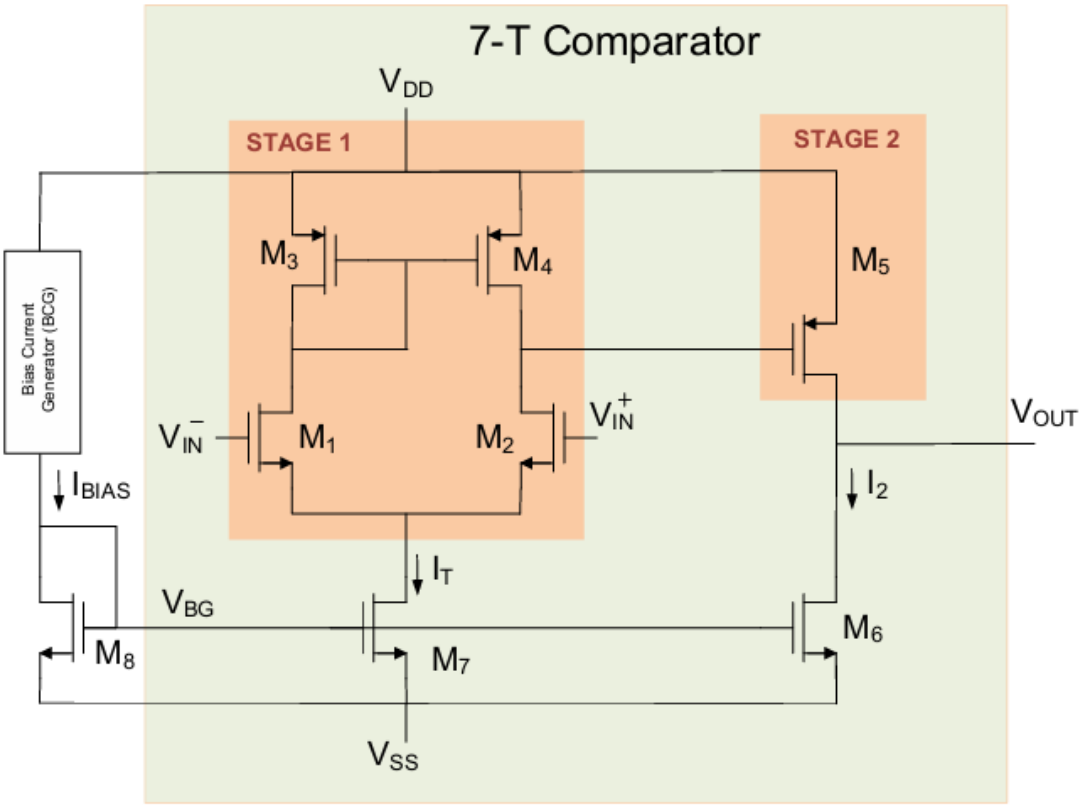
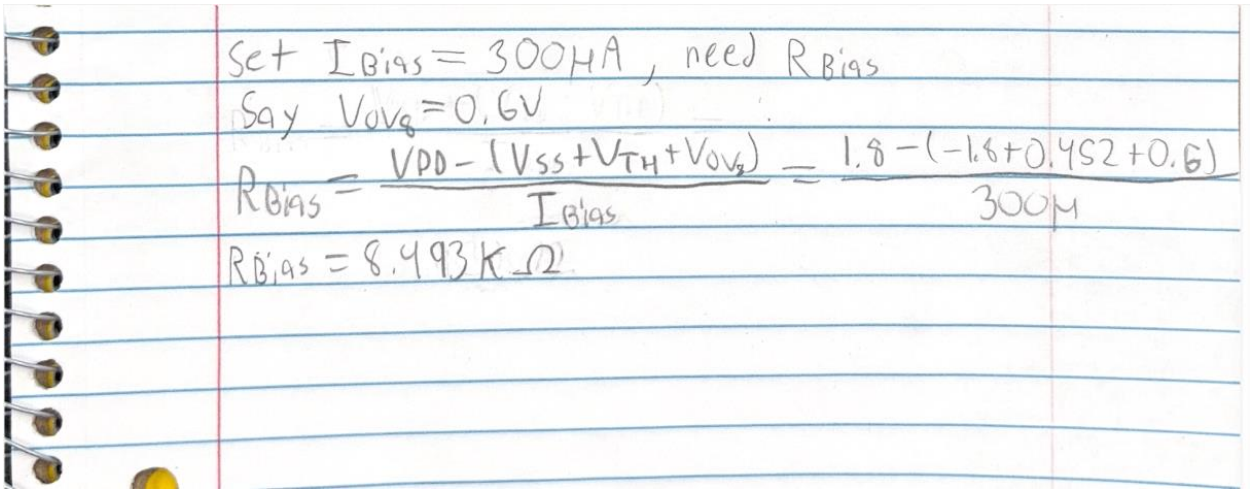



*Fig. 1, 7T amplifier schematic*

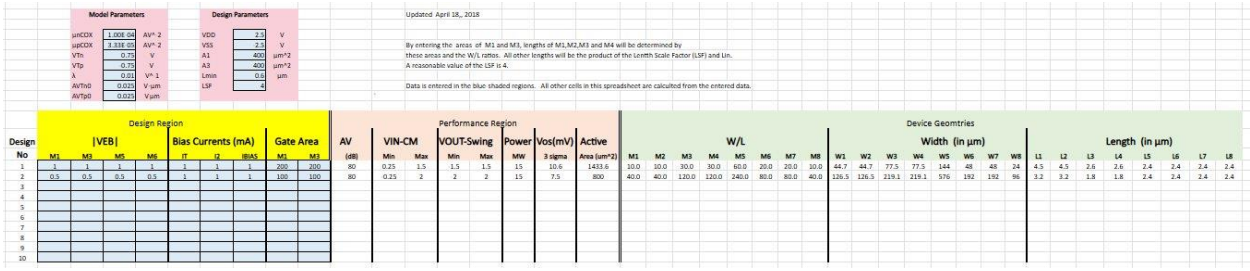
On top of the dimensions that we have to derive with the spreadsheet, we also have to set a bias current and then implement a bias current generator on the left side of the circuit. We decided that the best way to implement a bias current would be to use the VDD voltage already in the circuit and then a resistor to set the current. The calculation is given below.




*Fig*, 2, bias resistor calculation

I set the bias current to be 300uA and then the Vov8 to be 600mV. In the schematic, I could see that the source of the transistor below the resistor went to the -1.8V VSS. This means that gate voltage would be equivalent to VSS + VTH + VOV since overdrive and threshold voltage both take their share before reaching the source voltage node. Calculating the bias resistance got us a value of 8493 Ohms.

After calculating this, we moved on to finding the proper dimensions of the transistors. To start this, we had to open up the spreadsheet and start manipulating each variable in the blue shaded boxes.



Model Parameters

| | | |
|---|---|---|
| μnCOX | 1.00E-04 | A/V^2 |
| μpCOX | 3.33E-05 | A/V^2 |
| VTn | 0.75 | V |
| VTp | 0.75 | V |
| λ | 0.01 | V^-1 |
| AVTn0 | 0.025 | V·μm |
| AVTp0 | 0.025 | V·μm |

Design Parameters

| | | |
|---|---|---|
| VDD | 2.0 | V |
| VSS | 2.0 | V |
| A1 | 400 | μm^2 |
| A3 | 400 | μm^2 |
| Lmin | 0.6 | μm |
| LSF | 4 | |

Updated April 18,, 2018

By entering the areas of M1 and M3, lengths of M1,M2,M3 and M4 will be determined by these areas and the W/L ratios. All other lengths will be the product of the Lenth Scale Factor (LSF) and Lin. A reasonable value of the LSF is 4.

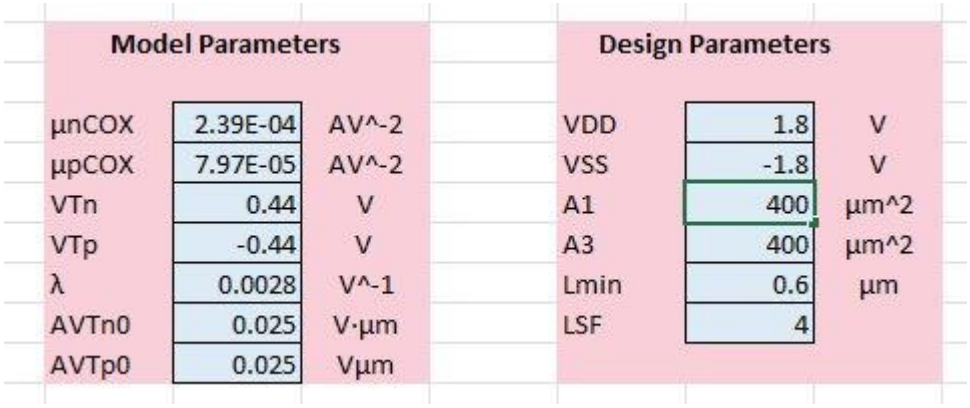
Data is entered in the blue shaded regions. All other cells in this spreadsheet are calculted from the entered data.

| Design No | Design Region \|VEB\| M1 | M3 | M5 | M6 | Bias Currents (mA) I1 | I2 | IBIAS | Gate Area M1 | M3 | AV (dB) | Performance Region VIN-CM Min | Max | VOUT-Swing Min | Max | Power MW | Vos(mV) 3 sigma | Active Area (um^2) | Device Geomtries W/L M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | Width (in μm) W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | Length (in μm) L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 200 | 200 | 80 | 0.25 | 1.5 | 1.5 | 1.5 | 15 | 10.6 | 1433.6 | 10.0 | 10.0 | 30.0 | 30.0 | 60.0 | 20.0 | 20.0 | 10.0 | 44.7 | 44.7 | 77.5 | 77.5 | 144 | 48 | 48 | 24 | 4.5 | 4.5 | 2.6 | 2.6 | 2.4 | 2.4 | 2.4 | 2.4 |
| 2 | 0.5 | 0.5 | 0.5 | 0.5 | 1 | 1 | 1 | 100 | 100 | 80 | 0.25 | 2 | 2 | 2 | 15 | 7.5 | 800 | 40.0 | 40.0 | 120.0 | 120.0 | 240.0 | 80.0 | 80.0 | 40.0 | 126.5 | 126.5 | 219.1 | 219.1 | 576 | 192 | 192 | 96 | 3.2 | 3.2 | 1.8 | 1.8 | 2.4 | 2.4 | 2.4 | 2.4 |
| 3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

*Fig. 3, spreadsheet view (untouched, doesn't contain our values)*

The first order was to obtain the model and design parameters of the NMOS2V transistor, which we have already done previously. This was completed in lab 7, so I went

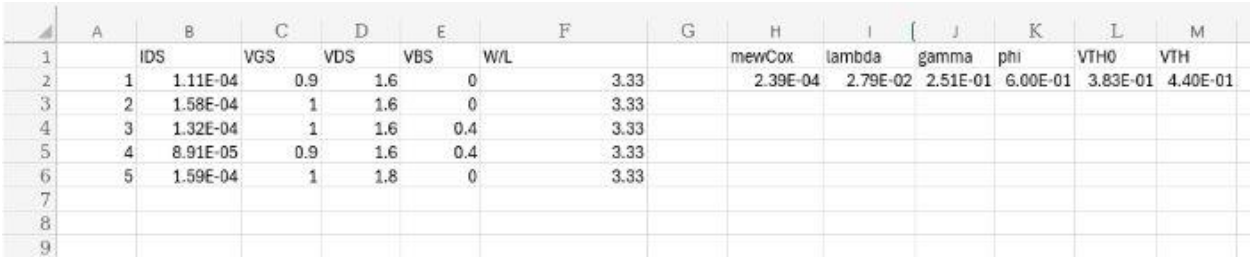
back to that lab report and filled in each of the parameters with our obtained values. The only values we didn't touch were the AVTn, AVTp, A1, A3, LSF, and Lmin.



| Model Parameters | | |
|---|---|---|
| μnCOX | 2.39E-04 | AV^-2 |
| μpCOX | 7.97E-05 | AV^-2 |
| VTn | 0.44 | V |
| VTp | -0.44 | V |
| λ | 0.0028 | V^-1 |
| AVTn0 | 0.025 | V·μm |
| AVTp0 | 0.025 | Vμm |

| Design Parameters | | |
|---|---|---|
| VDD | 1.8 | V |
| VSS | -1.8 | V |
| A1 | 400 | μm^2 |
| A3 | 400 | μm^2 |
| Lmin | 0.6 | μm |
| LSF | 4 | |

*Fig. 4, model and design parameters of spreadsheet*



| | A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | IDS | VGS | VDS | VBS | W/L | | mewCox | lambda | gamma | phi | VTH0 | VTH |
| 2 | 1 | 1.11E-04 | 0.9 | 1.6 | 0 | 3.33 | | 2.39E-04 | 2.79E-02 | 2.51E-01 | 6.00E-01 | 3.83E-01 | 4.40E-01 |
| 3 | 2 | 1.58E-04 | 1 | 1.6 | 0 | 3.33 | | | | | | | |
| 4 | 3 | 1.32E-04 | 1 | 1.6 | 0.4 | 3.33 | | | | | | | |
| 5 | 4 | 8.91E-05 | 0.9 | 1.6 | 0.4 | 3.33 | | | | | | | |
| 6 | 5 | 1.59E-04 | 1 | 1.8 | 0 | 3.33 | | | | | | | |
| 7 | | | | | | | | | | | | | |
| 8 | | | | | | | | | | | | | |
| 9 | | | | | | | | | | | | | |

*Fig. 5, parameters derived from lab 7*

After putting these in, we had to adjust VEB1 and VEB5 until we were able to achieve a gain of about 80dB on the spreadsheet. I also had to change the Ibias tab to 0.3mA since that's what I set the current to when calculating resistance. The VEB3 had to be adjusted to the mess with the size of transistors dimensions. We didn't touch IT and I2 since we heard from others that this could mess up the op amp easily. These 2 currents primarily affected the power output and some effect on the transistor dimensions. We set VEB6 around what the other VEBs were set to. Later on, we had to change VEB3 and the gate values in order to get rid of DRC errors.

| Design Region | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| \|VEB\| | | | | Bias Currents (mA) | | | Gate Area | |
| M1 | M3 | M5 | M6 | IT | I2 | IBIAS | M1 | M3 |
| 0.45 | 0.4 | 0.35 | 0.4 | 1 | 1 | 0.3 | 100 | 100 |
| 0.5 | 0.5 | | 0.5 | 1 | 1 | 1 | 100 | 100 |

*Fig. 6, design region values*

| Performance Region | | | | | | | |
|---|---|---|---|---|---|---|---|
| AV | VIN-CM | | VOUT-Swing | | Power | Vos(mV) | Active |
| (dB) | Min | Max | Min | Max | MW | 3 sigma | Area (um^2) |
| 80.20641495 | -0.498 | 1.4 | -1.4 | 1.45 | 10.8 | 15.5 | 411.3 |
| 80.20641495 | -0.448 | 1.3 | -1.3 | 1.8 | 10.8 | 7.5 | 800 |

*Fig. 7, final performance region values*

| Device Geomtries | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| W/L | | | | | | | | Width (in µm) | | | | | | | | Length (in µm) | | | | | | | | |
| M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 | |
| 21.4 | 21.4 | 81.2 | 81.2 | 212.2 | 54.2 | 54.2 | 27.1 | 46.3 | 46.3 | 90.1 | 90.1 | 38.199 | 9.7487 | 9.7487 | 4.8744 | 2.2 | 2.2 | 1.1 | 1.1 | 0.18 | 0.18 | 0.18 | 0.18 | |
| 17.3 | 17.3 | 52.0 | 52.0 | #NUM! | 34.7 | 34.7 | 17.3 | 83.3 | 83.3 | 144.2 | 144.2 | #NUM! | 6.23917 | 6.23917 | 3.1196 | 4.8 | 4.8 | 2.8 | 2.8 | 0.18 | 0.18 | 0.18 | 0.18 | |

*Fig. 8, Final device geometries*

Another important aspect of the op amp is that it meets all the performance requirements of the circuit. For this temperature sensor circuit, we had to go from 0V to 1V, so the output swing min had to be less than 0V and the max had to be more than 1V otherwise the output would get clipped before it reached its target. This design meets both of those criteria. The common mode voltage is also important to this circuit since the output will stop being linear if it goes beyond this limit. This design also meets those criteria since the greatest difference will be very small before it gets amplified. Power, Vos, and active were never mentioned as relevant by the TAs, so those were assumed to be in compliance with the circuit as well. After confirming all these values, we changed all the dimensions of the transistors to the values derived on the spreadsheet. The final part was to put all the pins on the circuit and turn it into a symbol so it could be tested for accuracy.

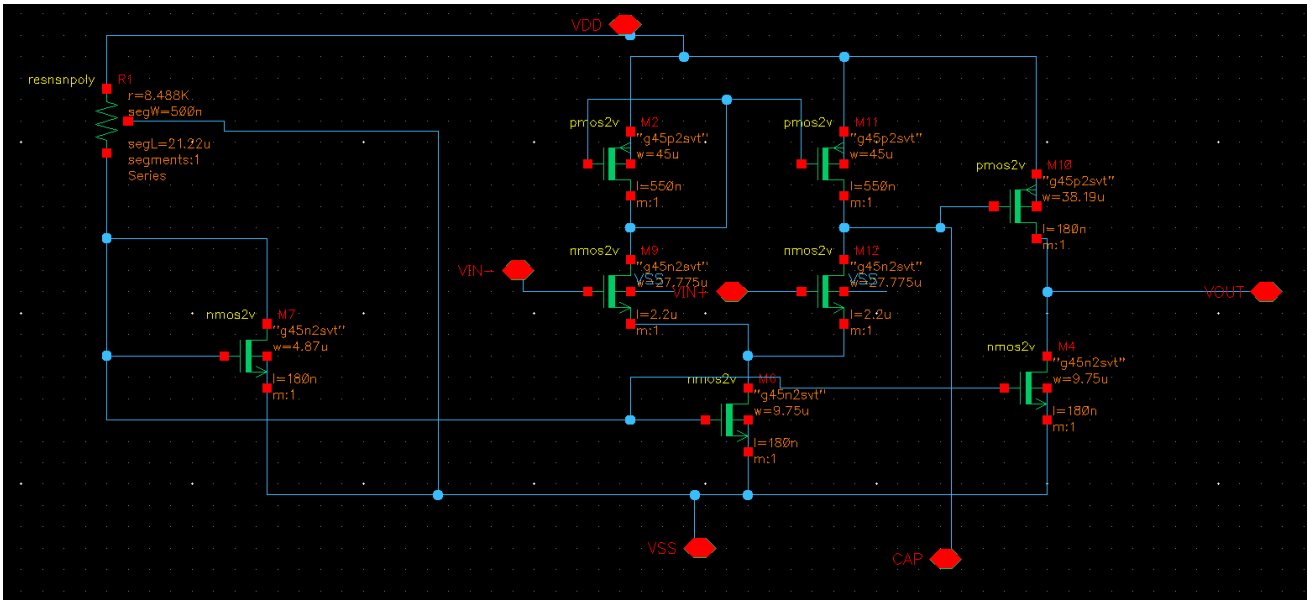



*Fig. 9, op amp circuit and values*

After turning it into a symbol, we had to create a testbench to ensure that the gain and phase margin were adequate for functionality. To do this, we took the symbol and gave it an AC source on the VIN+ terminal with a magnitude of 500mV. Next, we looped the VIN- terminal to the output VOUT to create an open loop gain so the dB could be measured. We also added the capacitors to the CAP and VOUT pins so that those wouldn't be shortened. The 1.8V VDD and -1.8V VSS were added for the op amp rails too. Finally, we placed a probe in the feedback line. This device will be crucial to getting the loop gain and phase margin.

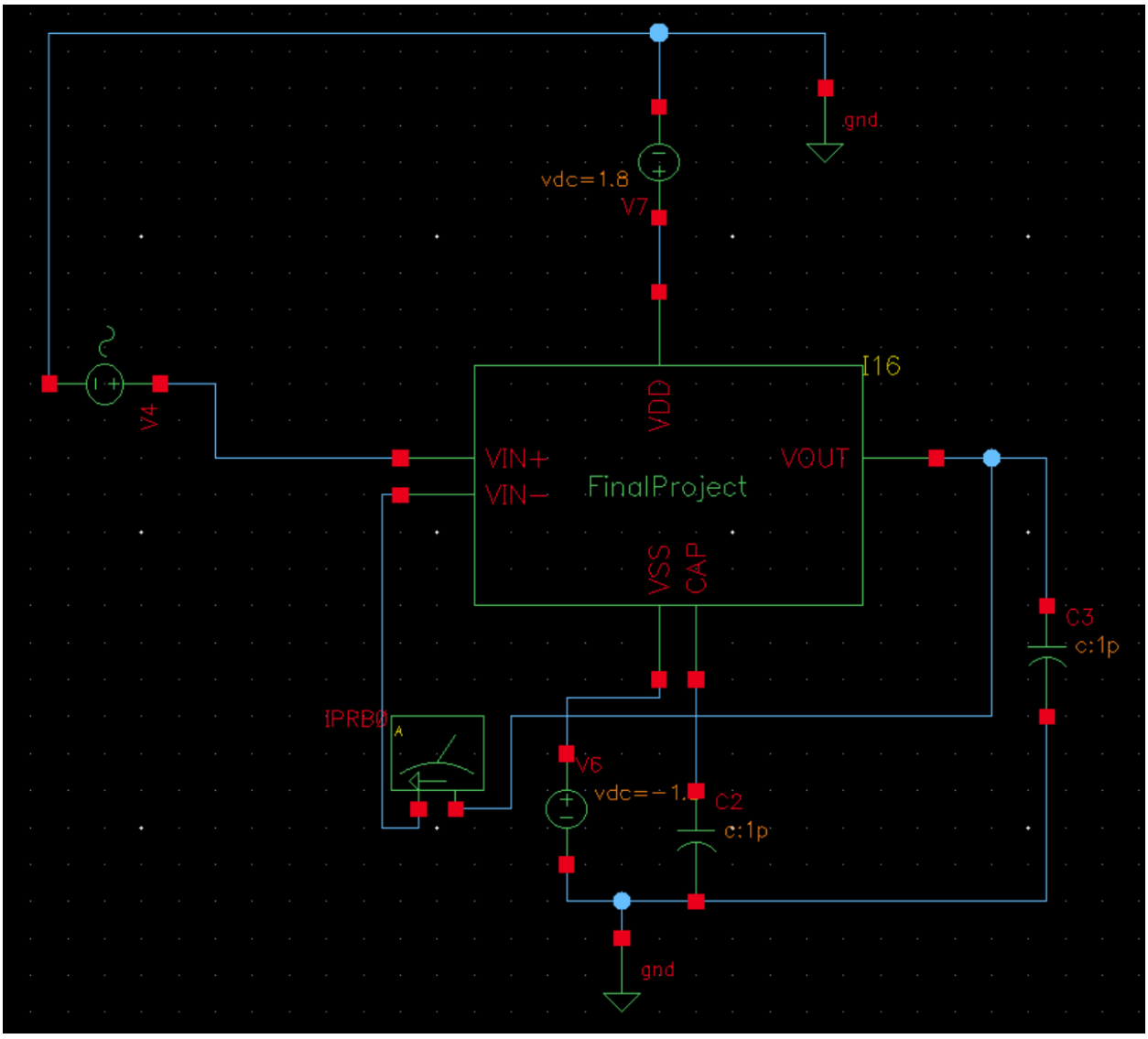



*Fig. 10, gain and phase margin testbench.*

In the ADE explorer, I set up the simulation for a DC analysis and a variable sweep of frequency to see the frequency response. I made the sweep logarithmic starting from 10Hz to 100MHz with points per decade of 20. Finally, I set the probe instance to the probe we placed in the loop and then made the VOUT line an output.

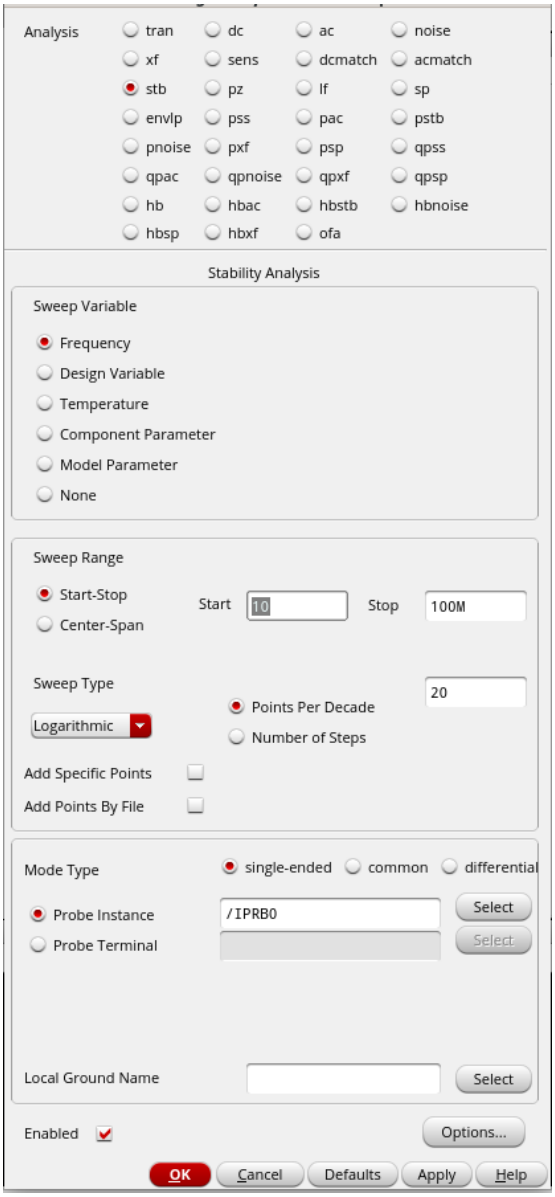



*Fig. 11, gain and phase margin simulation settings*

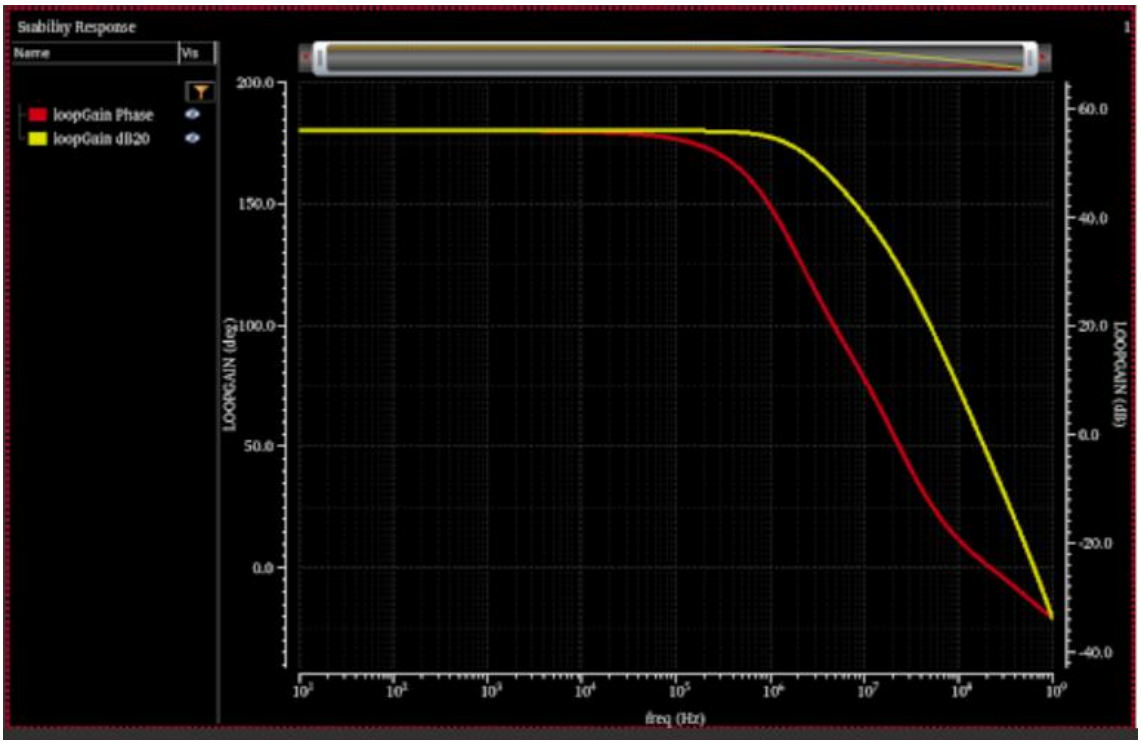



*Fig. 12, gain and phase margin plot*

After simulating, we found that the gain was approximately 53.8dB, which is quite far off from the 80dB that we expected to get. This could be caused by multiple different issues, such as the amplifier reaching its maximum potential gain or the transistor dimensions we edited to fit with the DRC and LVS didn't work effectively in the op amp. Even though this was decently far off from the desired value, the amplifier worked perfectly fine with only a little bit of gain error. To measure the phase margin, we had to find the angle of the phase where the gain was at 0dB. Looking at the graph, it can be seen that the phase is around 55 degrees at that point, meaning that the op amp is relatively stable and above the 45-degree minimum.

## Layout

For layout of the opamp this was a difficult task to complete. For reasons that will be apparent later, the layouts in both the high-level design and the opamp both had errors that the group was not able to successfully get rid of. To begin the layout, I made a layout design file and then brought over all of the components for the opamp to using the automated

layout design tool. This allows us to automatically have the components transfer over to layout, so we do not have to make the components again ourselves. Once we do this, it is just making all the connections asked of us from the layout design. This can be done by following the dashed line that is created when clicking on a new component or by following the schematic that was made prior to the layout. Once we were able to get the components to show up in layout, the team realized that the resistor and capacitors of the opamp did not transfer over. The reasoning that was not known to us until later was that we were using analoglib components and components from this library cannot be turned into layout. Because of this we had to find resistors and capacitors that would work. To get this to work, we were told by a fellow classmate (and was approved by a TA in the lab) that we could use 3-terminal resistors. These resistors are a part of the gpdk45 library and will allow us to have the resistors brought to layout. This third terminal for us did not show up in the layout and we were told that it wouldn't show up so we should be fine to use them. As far as the group knows this did not cause any errors to our knowledge for not having the bulk terminals of the resistors appear in the design of this opamp. Furthermore, we were originally told by a different student to use a MOSFET capacitor for the capacitor. This is a capacitor that was made to rearrange a MOSFET to act like one. This method did not work and caused many errors for us, so we decided to find a 3-terminal capacitor that was equivalent to how the resistor would act in layout. This helped us get the capacitor in our layout design of the opamp. This still caused frustrations to the design of our circuit layout. The reasoning is because when we incorporated it into the layout, the automatic designer messed up the connections of the capacitor, and we capacitor on how to add it to our layout. When discussing with other TAs and students that talked to the Dr. Geiger we were informed that Dr. Geiger is allowing the students to make the capacitor a pin and thus we can just add it as a part of the highest level design we have that would be used to test the output of the circuit. Because of this, we decided to change the capacitor to have a pin to make the layout easier for us. We named the capacitor pin ‘CAP’. From there we designed a reasonable layout that was ready for DRC checking.

When running the DRC check, we found that the circuit failed DRC. The reasoning was that a lot of my connections were wrong and needed to be fixed. At the time the group did not know about this until we talked to TA to confirm this. We later sat with a TA to fix all the errors that were being made on the DRC. When we finished the DRC errors, we were given a screen that looks like this through Pegasus:

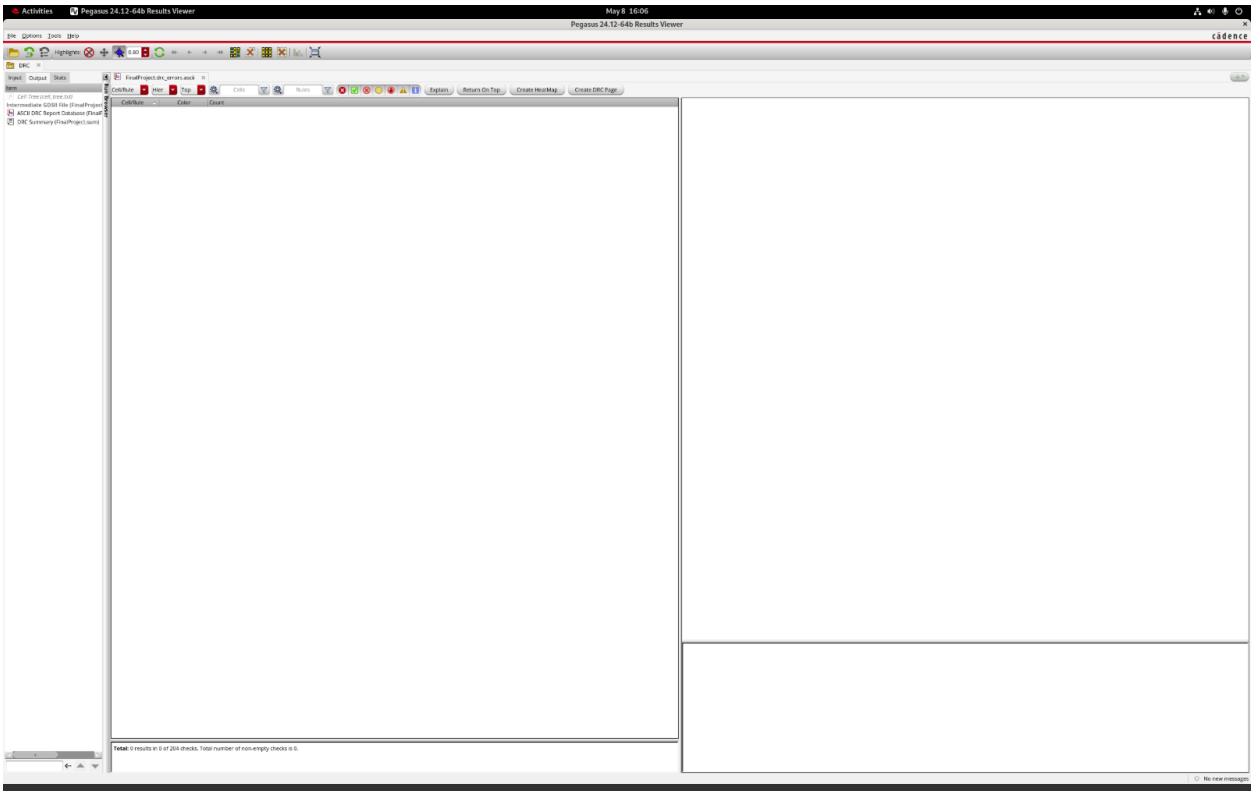



*Fig. 13, functional DRC check*

As we can see from the photo above, we ended up with no errors. This is a good sign because it shows that the DRC passed. Next up was to test using LVS. For LVS, we ran into many minor errors with spacing issues between metal connections. These issues were addressed and fixed promptly. The last errors that we had in our LVS was that the Layout transistors for the opamp had half of the widths for M1 and M3 and half the Lengths for M3 and M4. This issue was addressed by adjusting the widths and lengths of the transistors in the schematic to be similar to that of the layout design. However, because we are changing parameters of the transistors, this affects the output of the circuit which will in return cause us to make changes to our circuit to get the correct gain. We did this by messing with the components until we were given the correct output. Once we completed this task, we were still given the LVS error; however, the error was only read with a percentage less than 0.002. Due to this Sam, the TA allowed to keep these errors and move on to the high-level design. To end of this section, I will show the Layout of the opamp and the LVS error provided:

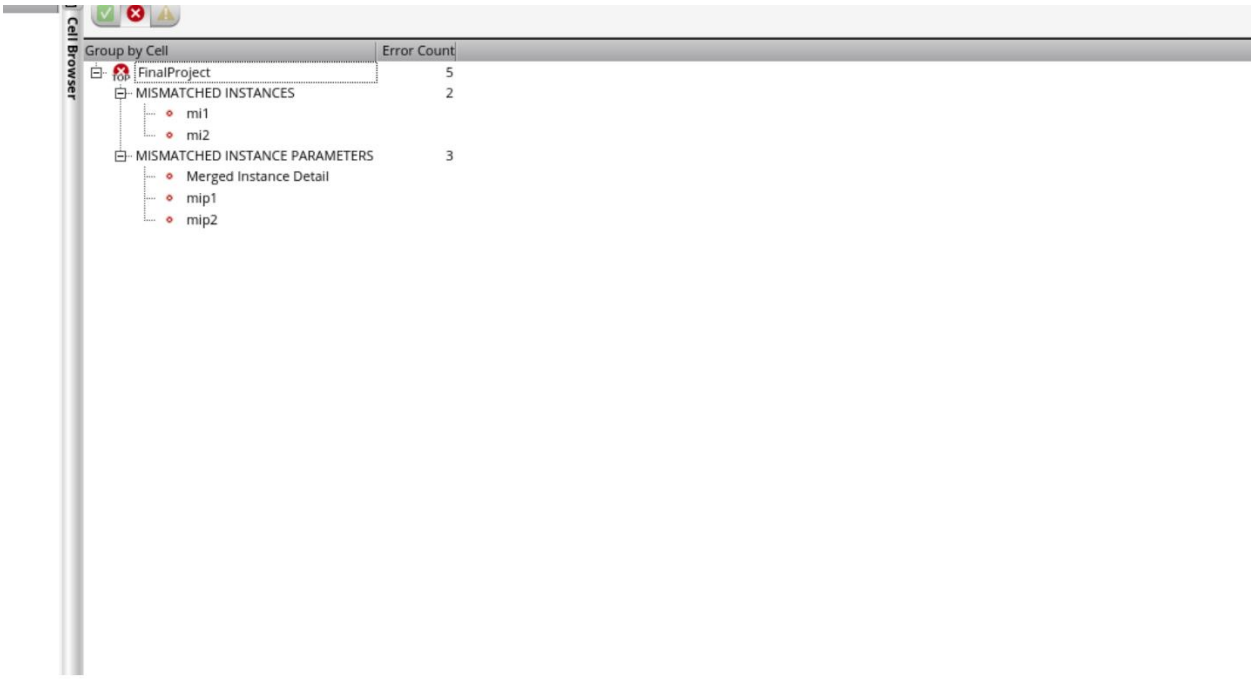



*Fig. 14, LVS check with very minor error percentage*

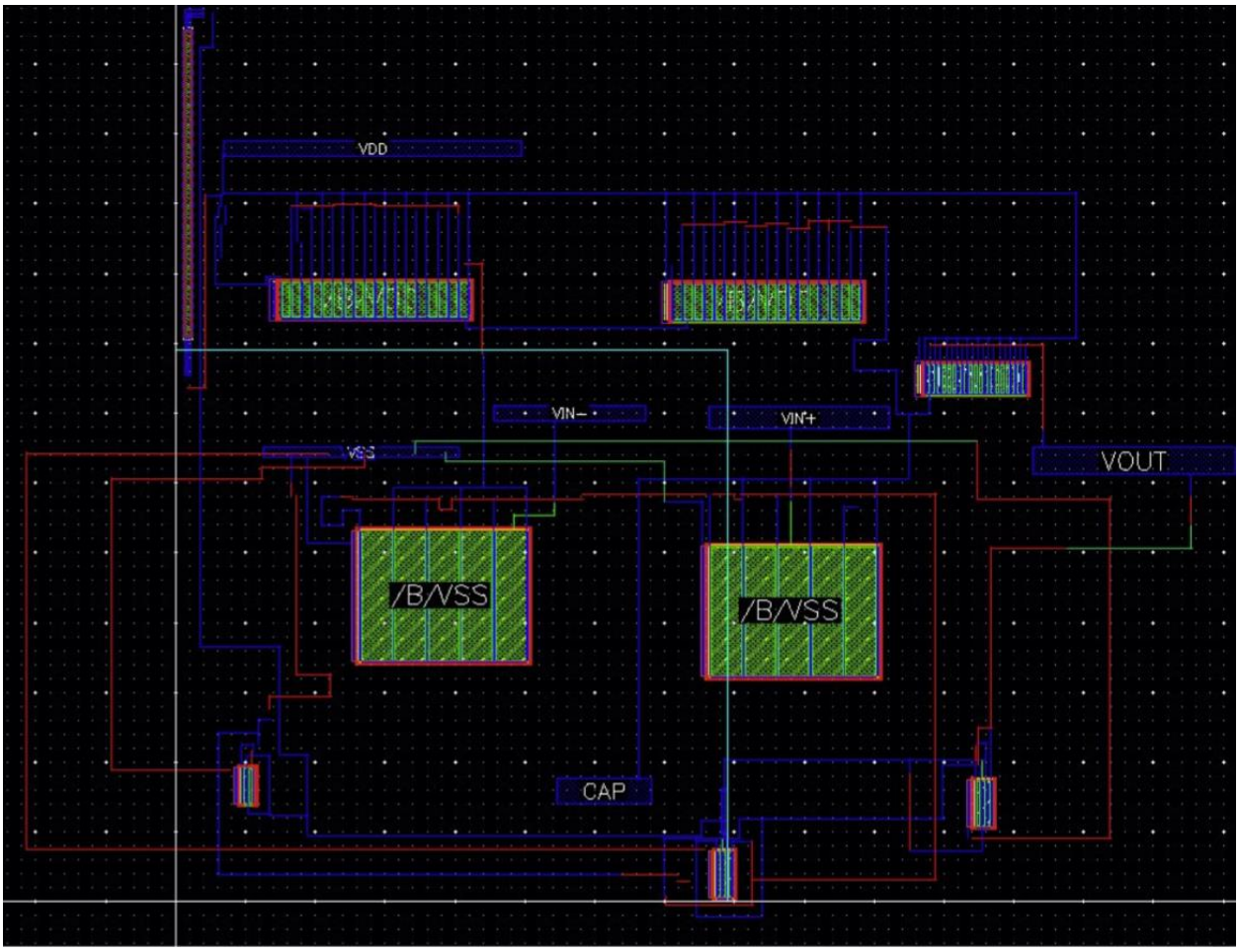


*Fig. 15, layout of op amp*

As you can see from the layout LVS, we only had errors from the square transistors in the middle of this layout photo above. It is because of the 0.002% error. As for the layout, we can see that we have the capacitor pin as Metal1 in the center of the layout. Although this is a rather imperfect design that would not get fabricated, it works well and we are happy with its results.

# High-Level Design

## Schematic

After completing the op amp, we had to move on to design the circuit for the actual temperature sensor. After talking to TAs about the potential design of the circuit, we found out that a PTAT temperature sensor circuit would be the perfect choice. PTAT stands for

proportional to absolute temperature, or in other words, as the temperature increases linearly on the circuit, the voltage at the output will also increase linearly. A PTAT is created by placing 2 differently sized diodes on each side of a differential amplifier terminal so that the difference in the transistors VBE values creates a reference voltage that will increase linearly. The reference voltage is directly affected by the ratio of the size of both diodes.

The first part of this circuit is the current mirror, which sets the current across both diodes in any ratio you want. I decided to connect VDD to the current mirror for simplicity and placed a resistor on the drain of the diode connected transistor to create a current to mirror. We had to make sure the resistance wasn't too large or there wouldn't be enough to saturate both diodes, so we chose to go with a value of 2KOhms. The left transistor went into the smaller sized diode while the right transistor went into a different resistor that connects to the larger diode. This resistor is used to set the PTAT currents effect on the reference voltage. Since this is not super important to find with set parameters, we just made this resistor 1KOhm and based calculations off that. As stated earlier, the reference voltage is equivalent to the differences in the voltage going across the 2 diodes. If you follow the node voltage equation of Vref you find that it is equal to VBE1 – VBE2, which will be an exponential subtracted by an exponential. This is what leads to the linearity of the output. After creating this design, we had to calculate the values of some of the other parameters.

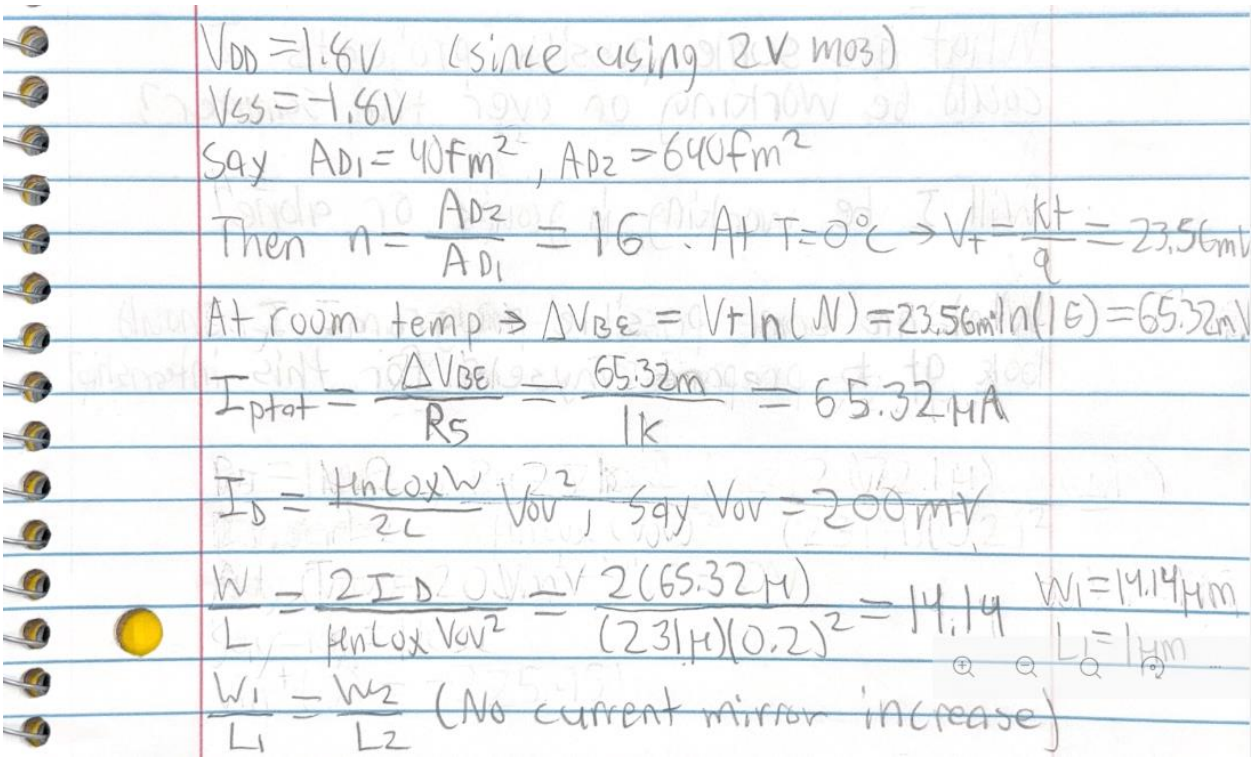



*Fig. 16, PTAT component calculations*

To start off the calculations, I set the area of diode 1 to 40fm^2 and diode to 2 640fm^2 and found a ratio of 16 between the 2. After this, I found the thermal voltage at 0C so it could be used in the difference in the VBE equation. After canceling and simplifying the 2 VBE equations, we found that the difference is just proportional to the ratio of the diodes. Using this, we calculated the differences in VBE at 0C. We could then find the PTAT current by dividing the difference over the resistor connected to the diode. I then set Vov for this pair of transistors as 200mV and solved to get a width and length ratio of 14.14um. We decided to make the current mirror 1:1, so both transistors had a width of 14.14um and a length of 1um.

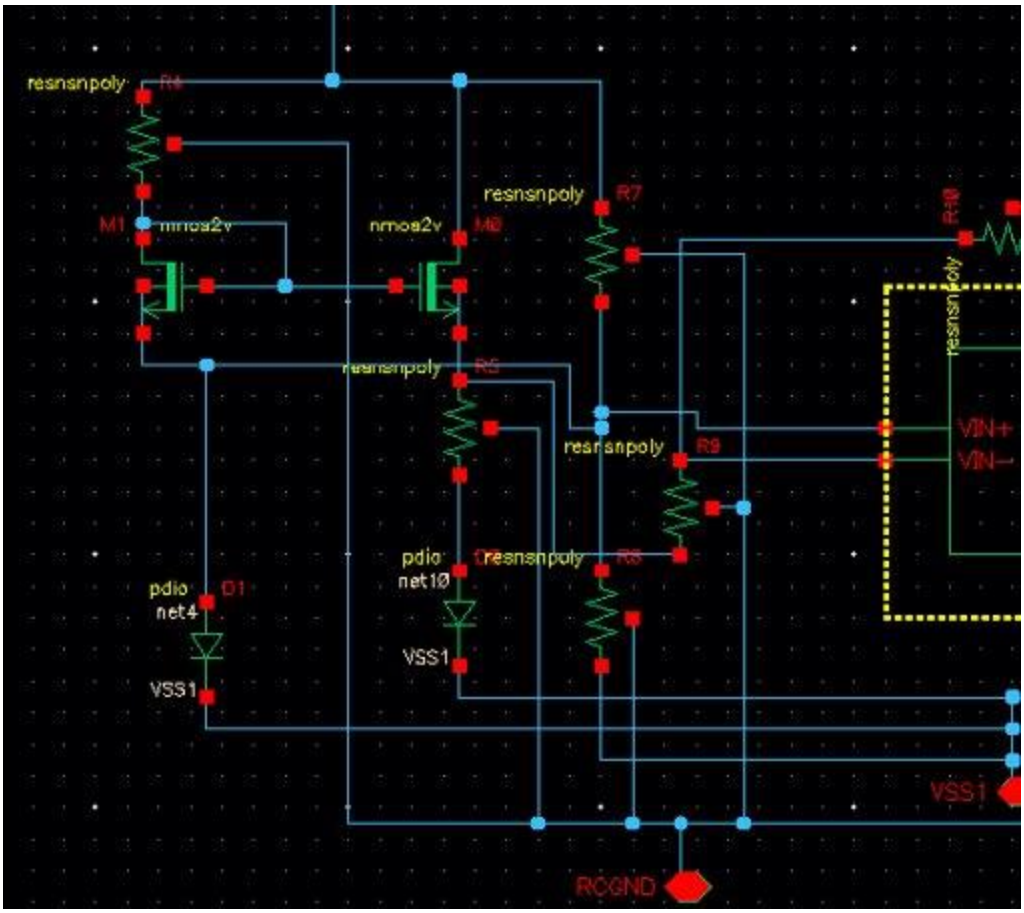



*Fig. 17, PTAT core of temp sensing circuit, note that top line connects to VDD*

After completing the PTAT side of the circuit, we had to focus on the offset created by the reference voltage. The reference voltage is located on the negative terminal of the amp, so initially at 0C the output will either be negative or railed to VSS if the offset is severe enough. To fix this offset, we can employ a voltage divider on the positive terminal of the op amp to add back what Vref took away at 0C. This will get us the desired starting point of 0C with some fine tuning of the divider. We didn't calculate anything for these parts since we just simulated the circuit until the output started at something close to 0V. The lower resistor, R8, we set at a constant 1KOhms while we adjusted the upper resistor, R7, however we needed to. We ended with a value of 1671.3 Ohms for R7.

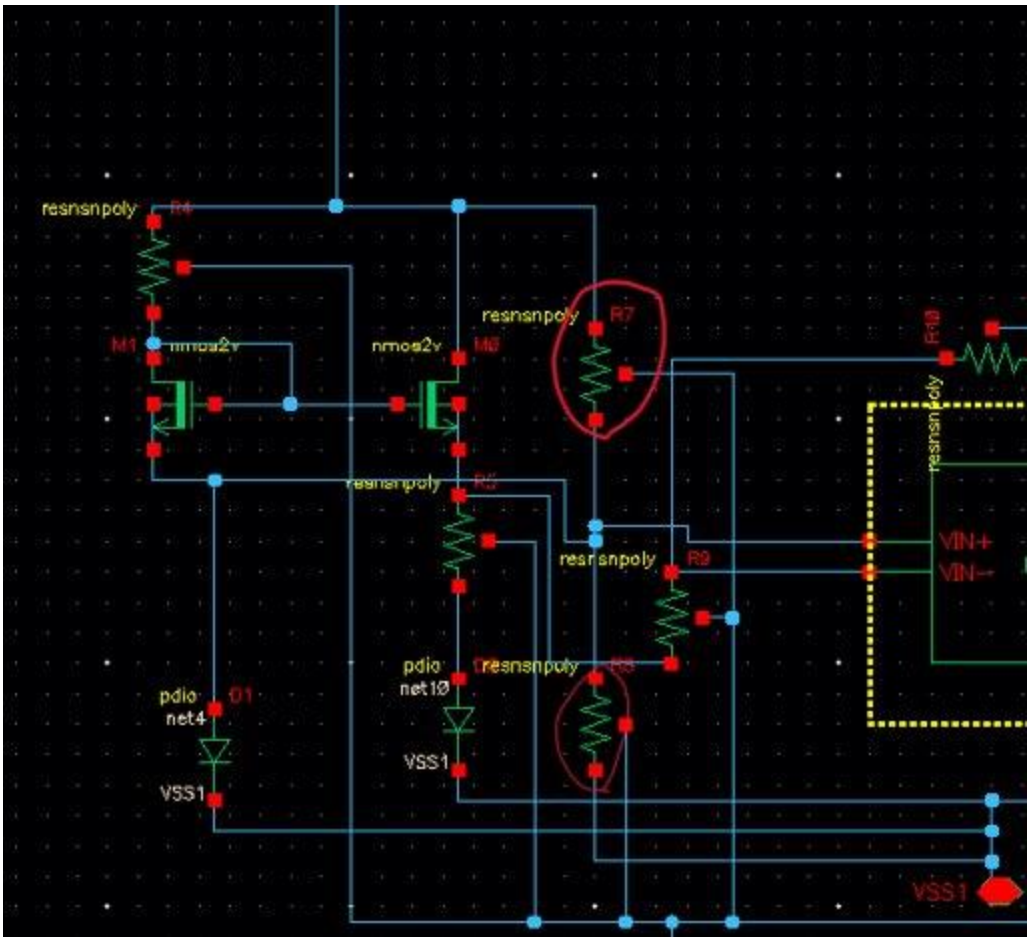



*Fig. 18, voltage divider configuration, resistors are circled in red*

The final issue to solve in this circuit was the small increase in Vref when temperature increased. We needed an increase of 20mV per degree to get to 1V, but the reference voltage only increased by less than 1mV at a time. This is where the gain section of the amplifier came in. To increase the gain, we needed a configuration on the negative terminal since that's where the reference voltage node is. This can be done by putting a resistor on the negative loop to Vout and one before the negative terminal node. This would form the gain equation of R10/R9. We set R9 to 1KOhm and played with R10 until we achieved a gain of about 20mV/C. After simulating with multiple different R10 values we found that a value of 47.2KOhm, which leads to a gain of 47.2V/V. The final thing to do was make all the inputs and outputs pins and turn this into a symbol for the final testbench.

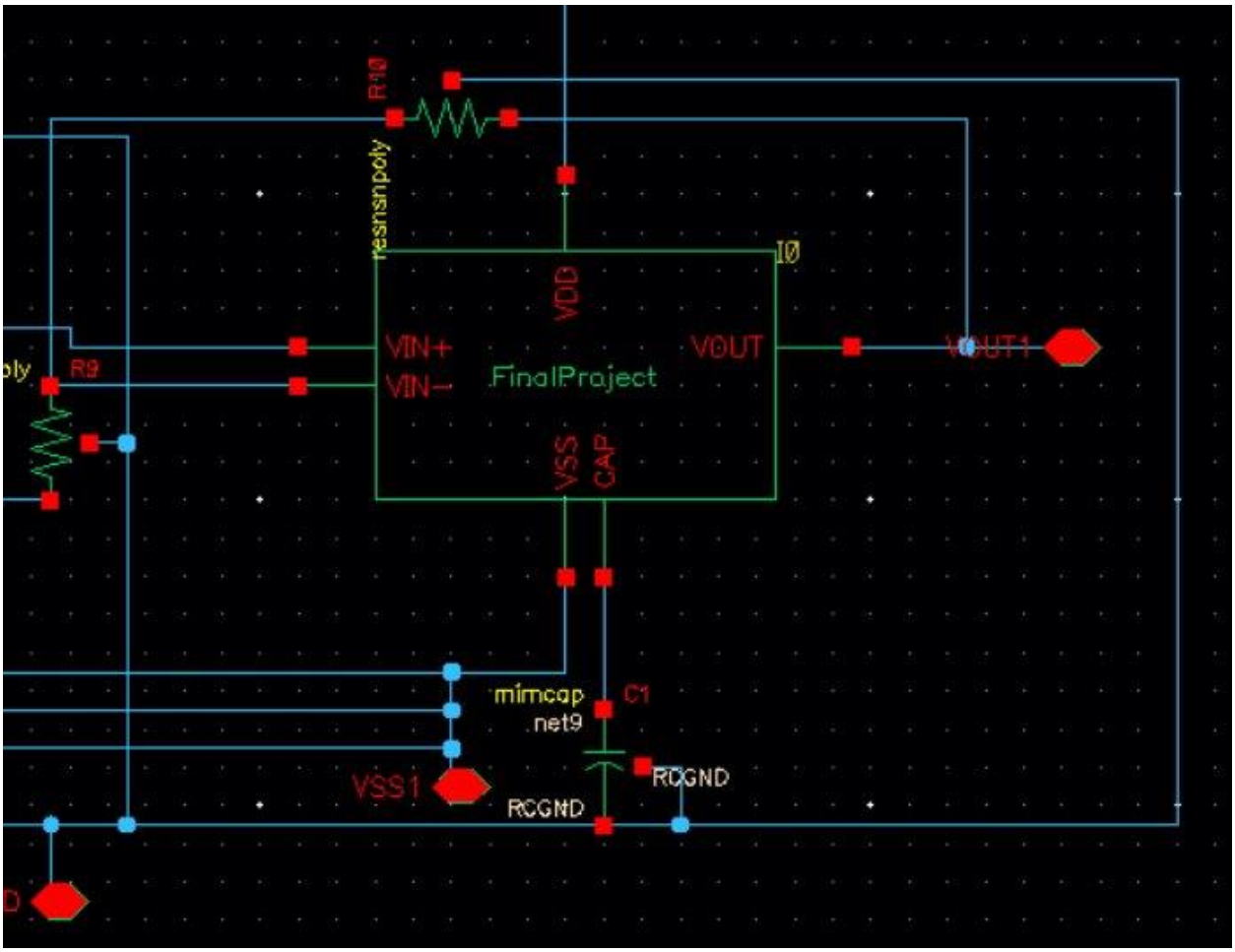



*Fig. 19, amplifier gain setup*

A quick note on the top 3 pictures is that these were taken before we changed a few small things on the circuit, but the design is the exact same as in our final schematic, and this one makes it a lot easier to see what's going on. Compared to the final circuit, we made the capacitor another pin for the final testbench and got rid of the RCGND pin. We then connected all the 3rd terminals of the resistors to the VSS pin. The final schematic is seen below.

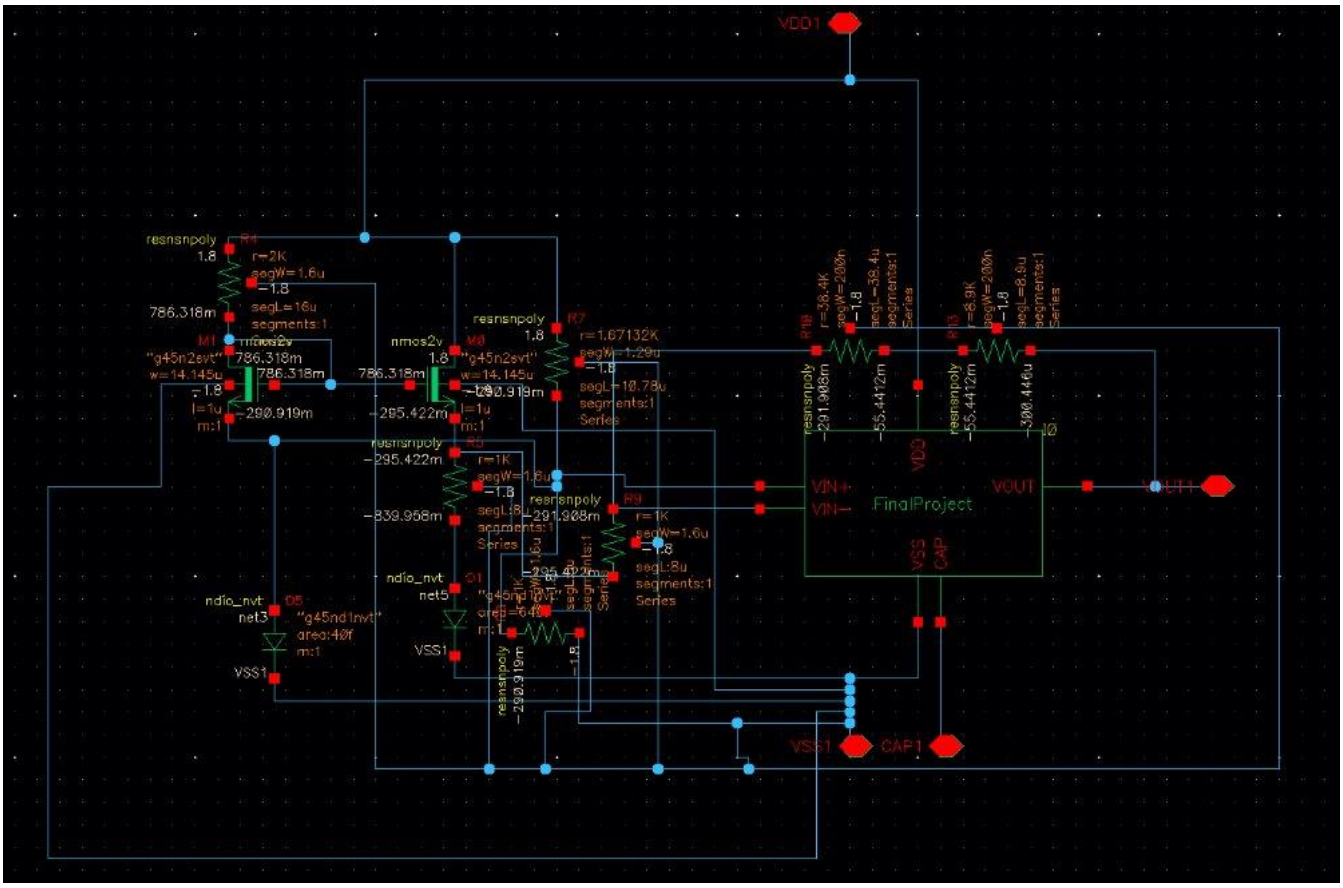



*Fig. 20, final temp sensor circuit*

After dialing all the values for the components and creating the pins, we could use the symbol to form the final testbench, which is seen below.

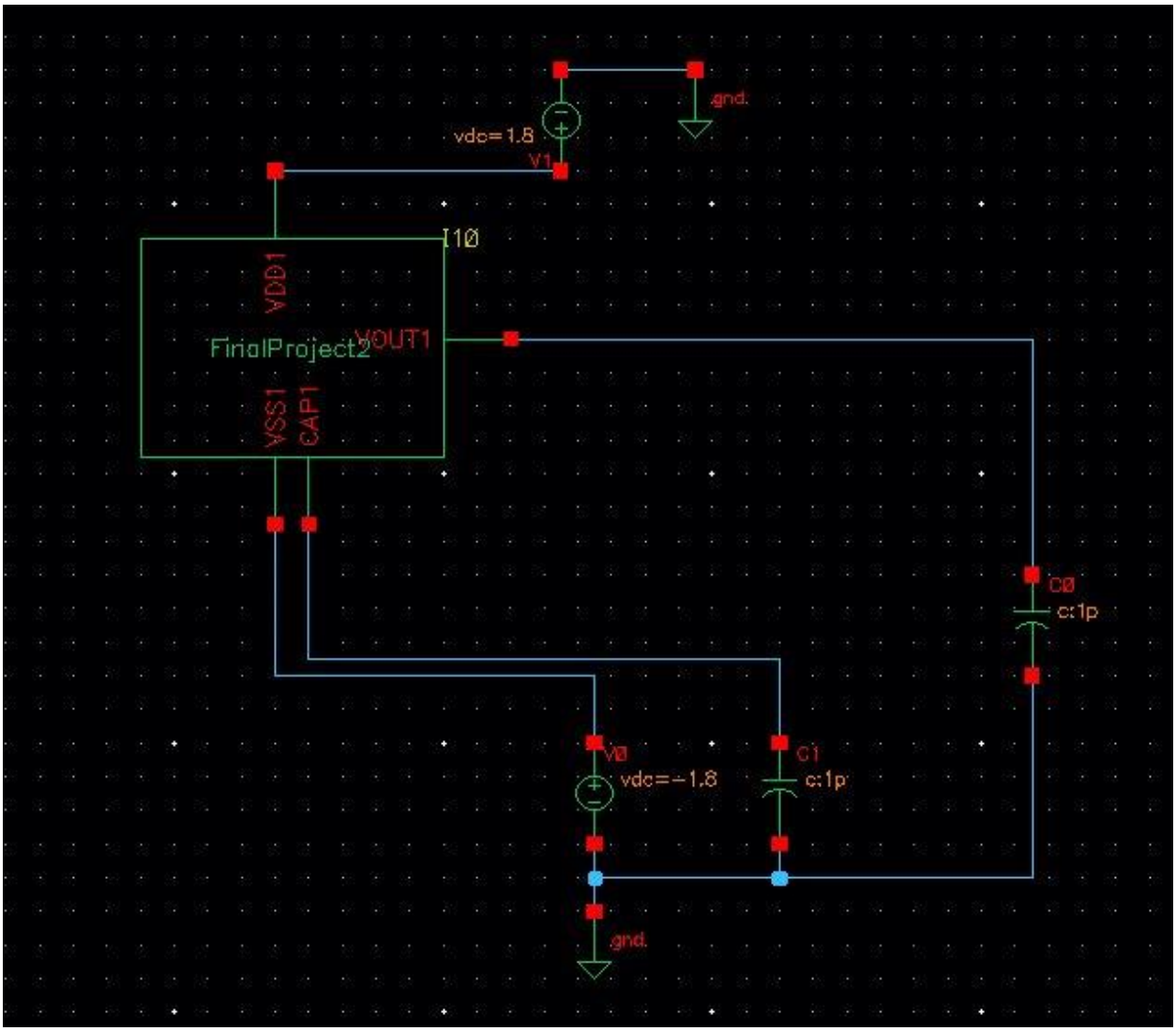



*Fig. 21, testbench of temp sensing circuit.*

As before, VDD was set to 1.8V and VSS was set to –1.8V. Both the CAP and output used ideal capacitors since this didn't need a layout made for it. To simulate this circuit, we used a DC sweep of temperature from 0C to 50C.

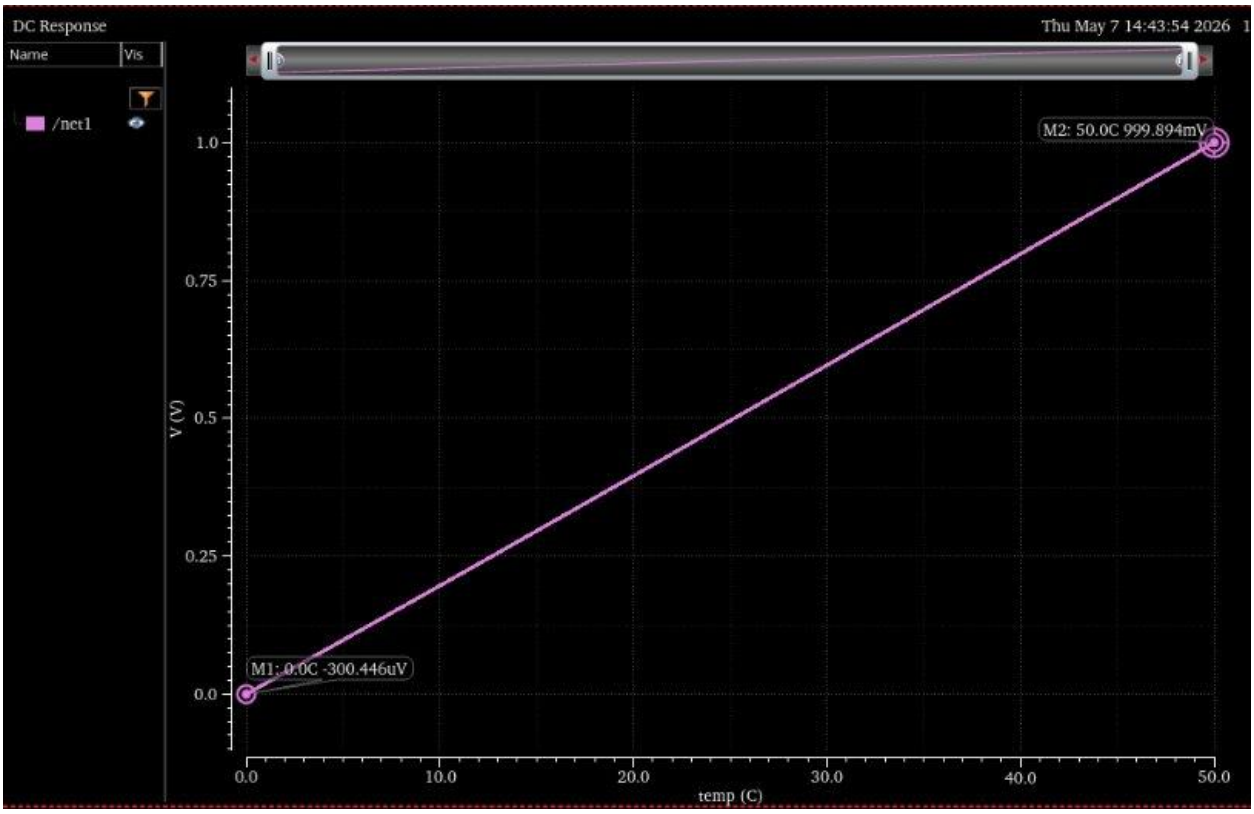


*Fig. 22, graph of final output*

After simulating we found that the circuit started very close to 0V, at about –300uV. This shows that we tuned our voltage divider very close to what Vref was at 0C. When measuring the total slope, we get a value of 20.004mV/C, which gets us extremely close to the desired 20mV/C with only a 0.02% error. We were also extremely close to 1V, only about 200uV short. The line is extremely linear showing that the PTAT did its job.

## Layout

Next is the layout for the High-level design. For the layout of the high-level design, this was a rigorous task to complete. To begin the layout, I made a layout design file and brought all the components for the design using the automated layout design tool. After this, I began to make all of the connections for the layout by using the same methods that were presented in the opamp layout subsection. Once I made all my connections, we tested the layout for the DRC. We made a lot of errors. Many were due to factors of the current mirror transistors not giving the proper widths and lengths and because of this we had to make more changes to the schematic. Not only this but our Diodes were not spaced out properly so because of

this we ran into the issue where we had to change the parameters of our diodes to get acceptable distance for the layout to not give this as an error. Once we were able to fix all these errors, we only had 5 remaining. The errors were due to p-substrate issues that when I talked to TAs about they all stated they could not figure out what the issue was and to either just take the points off the assignment or to use AI as a help tool. My lab partner and I both agreed to just take the points off given the fact that we had to study for finals. Next were the LVS errors. We had a lot of connection issue errors that when talking to the TA (Zak) he told us that we should not worry about the errors since it relates to an issue that would take us a long time to figure out.  We believe the errors in the LVS are rather basic errors that are out of our control to a degree. LVS checks love when the layout and the schematic are basically identical. When there is a slight variance, it can cause a lot of errors that are related to the situation. Eric and Connor spent a long time going through the layout and trying to match it with the schematic to confirm that there are no errors. After further review they came to the conclusion that nothing was wrong with their layout except for one thing. Although the layout and schematic are electronically the same, all the connections are there and correct. However, the layout differs from the schematic in the sense that the connects are wired in a different orientation compared to the layout. We believe this is the root of the issue. After conversing with Sam, we just decided to accept the points off even though we still think our layout is correct. Here is a photo of the DRC and LVS for the layout as well as a photo of the entire layout. Here are the photos:

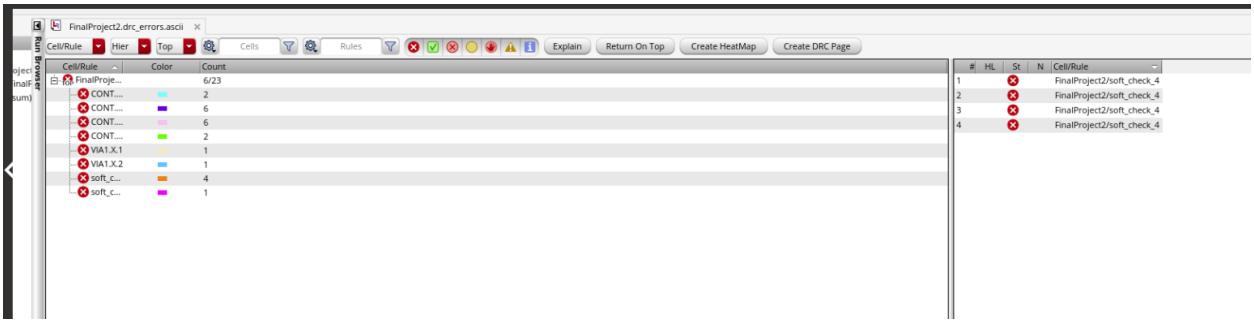



*Fig. 23, DRC check*

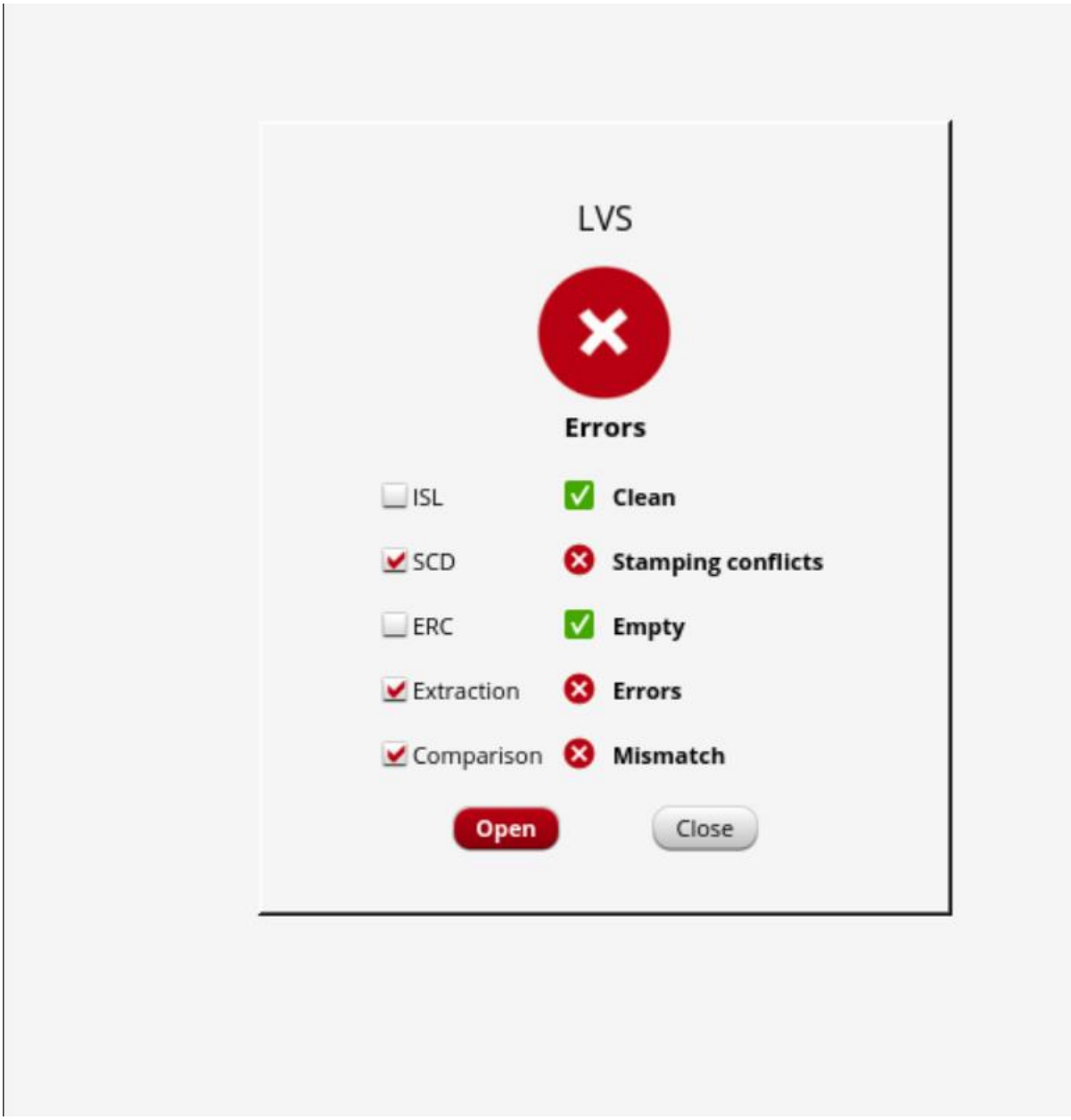



*Fig. 24, LVS check*

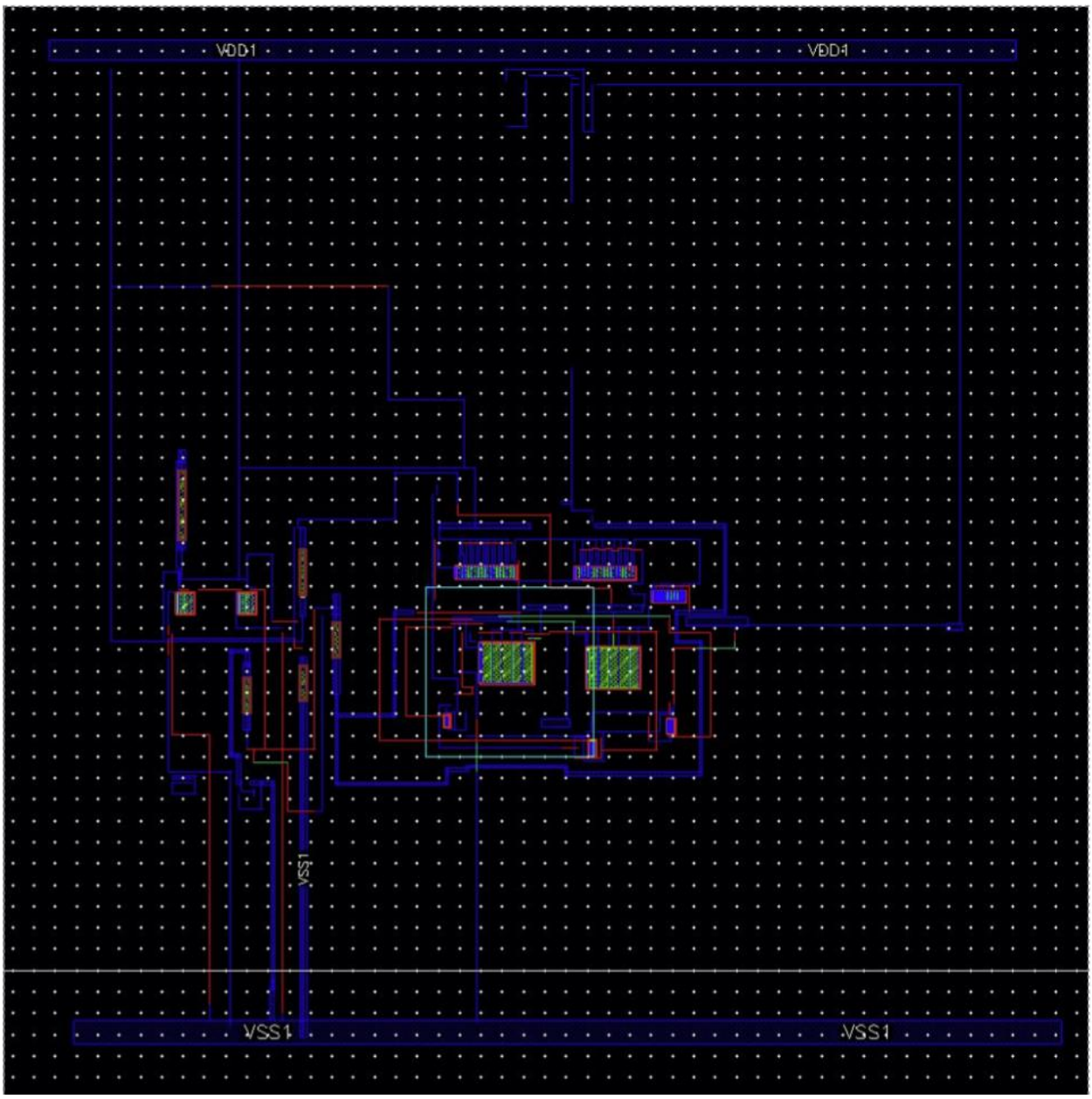



*Fig. 25, layout of whole temp sensor circuit*

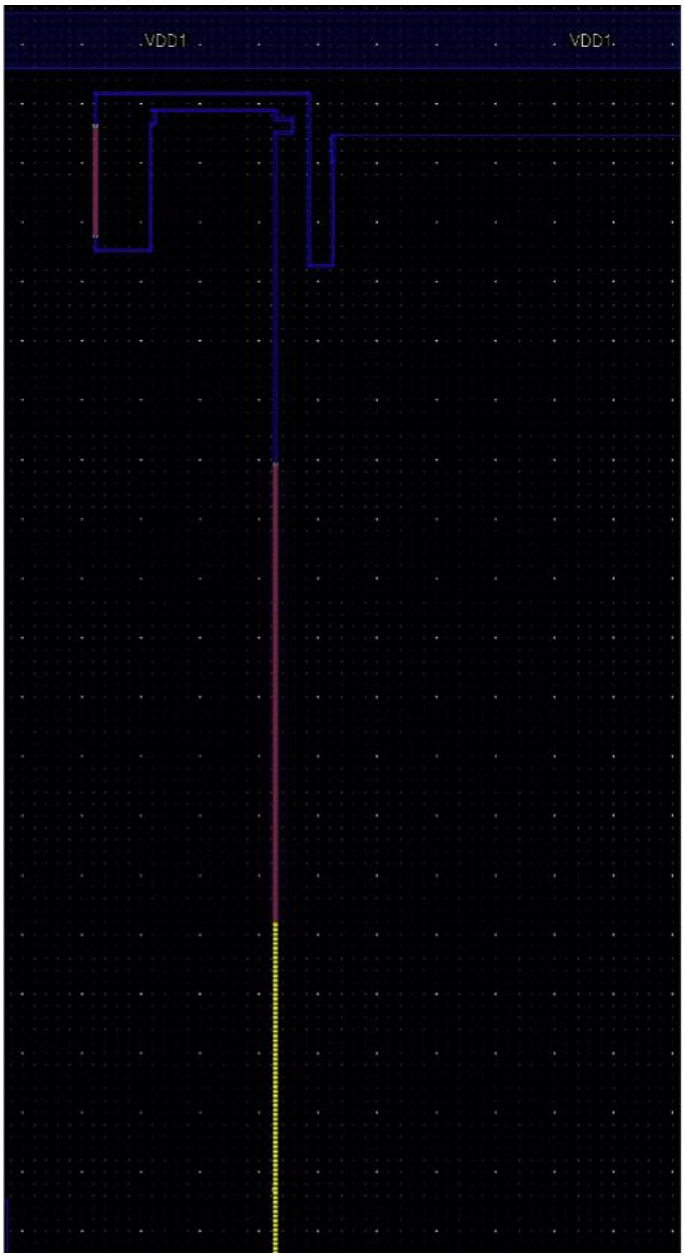



*Fig. 26, of resistors in viewable magnefication*

Resistors that are hard to see at the top that look like they are missing

As the reader can see, LVS and DRC had errors, and we still think that these are errors that are fine to have in a 330 project since we feel that a lot of it is outside the scope of our knowledge. However, we do understand the reason we are getting the points off. In terms of the layout itself the photo provided above gives enough justice to fully grasp the concept of our Temperature sensor however there are parts to it that look like they are

missing because of how small they are  which are the two resistors at the top of the page. Here above his paragraph is a photo of the resistor as proof that they exist it is just that they are small.

## Conclusion

Overall for this project was a fun experience to have. We have both made Temperature Sensors in the past for EE230, so getting to expand on that project was great for us to fully understand it. The structure of how the lab was designed for us to build an opamp when we barely covered the design of opamps was bit of a struggle. However, all the lab projects were fun and enjoyable. Even though EE330 is technically an optional course, it was an especially useful course to take students interested in understanding circuit theory in greater detail. I look forward to how EE435 will treat us in the future given that it is another course taught by Dr. Geiger and the follow up to EE 330.